

Report

Node description

Model name:	Intel(R) Xeon(R) Gold 6248 CPU @ 2.50GHz
CPU(s):	80
Server name:	ProLiant XL270d Gen10
NUMA node(s):	2
NUMA node0 CPU(s):	0-19,40-59
NUMA node1 CPU(s):	20-39,60-79
node 0 size:	385636 MB
node 0 free:	105947 MB
node 1 size:	387008 MB
node 1 free:	232013 MB
OS:	Ubuntu 22.04.5 LTS

Scaling

Interval	Number of steps	Threads														
		1	2		4		7		8		16		20		40	
		T(1), s	T(2), s	S(2)	T(4), s	S(4)	T(7), s	S(7)	T(8), s	S(8)	T(16), s	S(16)	T(20), s	S(20)	T(40), s	S(40)
[-2, 2]	40 000 000	~0.46	~0.24	~1.91	~0.13	~3.58	~0.079	~5.92	~0.071	~6.41	~0.041	~11.24	~0.036	~12.83	~0.025	~18.36

Conclusion on thread usage

As we can see on the graph below, some small usage of threads can result in performance close to linear. But by increasing the number of threads, the relationship ends up being from slightly worse to the point, where the results are extremely far from linear values.

